SOLUTIONS FOR WORKSHEET 4

Question 1. Prove that the following series are divergent.

(a) $\sum_{k=1}^{\infty}(-1)^k$ (b) $\sum_{k=1}^{\infty}\frac{2k+1}{k+3}$ (c) $\sum_{k=1}^{\infty}\frac{1}{2}$.

Solution. Divergence test (DT): $\lim\left(a_k\right)\neq 0$ or does not exist $\Rightarrow \sum_{k=1}^{\infty}a_k$ diverges.

Geometry series (GS): $\sum_{k=1}^{\infty}r^k$ converges $\Leftrightarrow \left|r\right|<1$.

(a) $\lim\left((-1)^k\right)$ does not exist $\Rightarrow \sum_{k=1}^{\infty}a_k$ diverges by DT.

(b) $\lim\left(\frac{2k+1}{k+3}\right)=2\neq 0\Rightarrow \sum_{k=1}^{\infty}a_k$ diverges by DT.

(c) $\lim\left(\frac{1}{2}\right)=\frac{1}{2}\neq 0\Rightarrow \sum_{k=1}^{\infty}a_k$ diverges by DT.

Question 2. Discuss the convergence of the following series.

(a) $\sum_{k=1}^{\infty}\frac{\cos(k)}{k^2+1}$ (b) $\sum_{k=1}^{\infty}\frac{k^2\sin(k)}{k^4+3k^2+1}$ (c) $\sum_{k=1}^{\infty}\frac{(-1)^k k^2}{2^k}$

Solution. p-series test (PST): $\sum_{k=1}^{\infty}\frac{1}{k^p}$ converges if $p>1$; and diverges if $p\leq 1$.

Comparison test (CT): If $0\leq a_k\leq b_k\,\forall k\in\mathbb{N}$, then $\sum_{k=1}^{\infty}b_k$ converges $\Rightarrow \sum_{k=1}^{\infty}a_k$ converges.

Absolute convergence test (ACT): $\sum_{k=1}^{\infty}\left|a_k\right|$ converges $\Rightarrow \sum_{k=1}^{\infty}a_k$ converges.

(a) $\left|\frac{\cos(k)}{k^2+1}\right|\leq\frac{1}{k^2+1}\leq\frac{1}{k^2}$ ($p=2>1$, $\sum_{k=1}^{\infty}\frac{1}{k^2}$ converges). By CT, $\sum_{k=1}^{\infty}\left|\frac{\cos(k)}{k^2+1}\right|$ converges and by ACT, $\sum_{k=1}^{\infty}\frac{\cos(k)}{k^2+1}$ converges.

(b) $\left|\frac{k^2\sin(k)}{k^4+3k^2+1}\right|\leq\frac{k^2}{k^4+3k^2+1}\leq\frac{k^2}{k^4}=\frac{1}{k^2}$ ($\sum_{k=1}^{\infty}\frac{1}{k^2}$ converges). By CT, $\sum_{k=1}^{\infty}\frac{k^2\sin(k)}{k^4+3k^2+1}$ converges and by ACT, $\sum_{k=1}^{\infty}\frac{k^2\sin(k)}{k^4+3k^2+1}$ converges.

(c) $\left|\frac{\frac{(k+1)^2}{2^{k+1}}}{\frac{k^2}{2^k}}\right|=\frac{1}{2}\left(\frac{k+1}{k}\right)^2\to\frac{1}{2}<1$ as $k\to\infty$. So, $\sum_{k=1}^{\infty}\left|\frac{(-1)^k k^2}{2^k}\right|$ converges and hence $\sum_{k=1}^{\infty}\frac{(-1)^k k^2}{2^k}$ converges by ACT.

Question 3. Using the ratio test, discuss the convergence of the following series.

(a) $\sum_{k=1}^{\infty} \frac{6^k}{k!}$ (b) $\sum_{k=1}^{\infty} \frac{k3^k}{4^k}$ (c) $\sum_{k=1}^{\infty} \frac{5^k}{4^k k}$ (d) $\sum_{k=1}^{\infty} \frac{1}{k}$.

Solution. Ratio Test (RT): Suppose $a_k > 0$ for all $k \geq 1$, and $\lim\left(\frac{a_{k+1}}{a_k}\right) = L$. Then the series $\sum_{k=1}^{\infty} a_k$ converges if $L < 1$ and diverges if $L > 1$. Note that if $L = 1$, the result is inconclusive.

(a) Since $\lim\left(\frac{\frac{6^{k+1}}{(k+1)!}}{\frac{6^k}{k!}}\right) = \lim\left(\frac{6}{k+1}\right) = 0 < 1$, it follows by RT that the series converges.

(b) Since $\lim\left(\frac{\frac{(k+1)3^{k+1}}{4^{k+1}}}{\frac{k3^k}{4^k}}\right) = \lim\left(\frac{3(k+1)}{4k}\right) = \frac{3}{4} < 1$, it follows by RT that the series converges.

(c) Since $\lim\left(\frac{\frac{5^{k+1}}{4^{k+1}(k+1)}}{\frac{5^k}{4^k k}}\right) = \lim\left(\frac{5k}{4(k+1)}\right) = \frac{5}{4} > 1$, it follows by RT that the series diverges.

(d) $\lim\left(\frac{\frac{1}{k+1}}{\frac{1}{k}}\right) = \lim\left(\frac{k}{k+1}\right) = 1$, so we cannot obtain any conclusion using RT.

Alternative. n-th root test (NRT): Suppose $a_n > 0$ for all $n \geq 1$, and $\lim \sqrt[n]{a_n} = L$. Then the series $\sum_{n=1}^{\infty} a_n$ converges if $L < 1$ and diverges if $L > 1$. Note that if $L = 1$, the result is inconclusive. Also note that $\lim\left(\sqrt[n]{n}\right) = 1$.

(b) Since $\lim\left(\sqrt[k]{\frac{k3^k}{4^k}}\right) = \frac{3}{4}\lim\left(\sqrt[k]{k}\right) = \frac{3}{4} < 1$, it follows that $\sum_{k=1}^{\infty} \frac{k3^k}{4^k}$ converges.

(c) Since $\lim\left(\sqrt[k]{\frac{5^k}{k4^k}}\right) = \frac{5}{4}\lim\left(\frac{1}{\sqrt[k]{k}}\right) = \frac{5}{4} > 1$, it follows that $\sum_{k=1}^{\infty} \frac{k3^k}{4^k}$ converges.

Proof. Note that $\left(1+x\right)^n = 1 + nx + \frac{n(n-1)}{2}x^2 + \ldots + x^n \geq \frac{n(n-1)}{2}x^2 \ \forall x \geq 0$.

Let $y = 1 + x \geq 1$. Then $y^n \geq \frac{n(n-1)}{2}(y-1)^2 \geq \frac{n^2}{4}(y-1)^2$ because $n - 1 \geq \frac{n}{2} \forall n \geq 1$.

Let $y = \sqrt[n]{n}$, then $n \geq \frac{n^2}{4}\left(\sqrt[n]{n} - 1\right)^2 \Rightarrow \sqrt{n} \geq \frac{n}{2}\left(\sqrt[n]{n} - 1\right) \Rightarrow \frac{2}{\sqrt{n}} \geq \sqrt[n]{n} - 1$.

By Archimedean property: $\forall \varepsilon > 0 \ \exists \sqrt{N} \in \mathbb{N} \ni \frac{\varepsilon}{2} > \frac{1}{\sqrt{N}}$ (N is a perfect square).

Thus, $n > N \Rightarrow \left|\sqrt[n]{n} - 1\right| = \sqrt[n]{n} - 1 \leq \frac{2}{\sqrt{n}} < \frac{2}{\sqrt{N}} < \varepsilon$.

Question 4. Using the p-series test, discuss the convergence of the following series.

(a) $\sum_{k=1}^{\infty} \frac{k^2}{k^3+1}$ (b) $\sum_{k=1}^{\infty} \frac{k}{k^3+1}$ (c) $\sum_{k=1}^{\infty} \frac{(k+1)\sin^2 k}{k^3+3k+1}$.

Solution. p-series test: $\sum_{k=1}^{\infty} \frac{1}{k^p}$ converges if $p > 1$; and diverges if $p \le 1$.

(a) For $k \ge 1$, $\frac{k^2}{k^3+1} \ge \frac{k^2}{k^3+k^3} = \frac{1}{2k} > 0$ and since $\sum_{k=1}^{\infty} \frac{1}{k}$ diverges ($p \le 1$), it follows from CT that $\sum_{k=1}^{\infty} \frac{k^2}{k^3+1}$ is divergent.

(b) For $k \ge 1$, $0 < \frac{k}{k^3+1} \le \frac{k}{k^3} = \frac{1}{k^2}$ and since $\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges ($p = 2 > 1$), it follows from CT that $\sum_{k=1}^{\infty} \frac{k}{k^3+1}$ is convergent.

(c) For $k \ge 1$, $0 < \frac{(k+1)\sin^2 k}{k^3+3k+1} \le \frac{k+1}{k^3+3k+1} \le \frac{k+k}{k^3} = \frac{2}{k^2}$ and since $2\sum_{r=1}^{\infty} \frac{1}{r^2}$ converges ($p = 2 > 1$ and Theorem 3.5), it follows from CT that $\sum_{k=1}^{\infty} \frac{(k+1)\sin^2 k}{k^3+3k+1}$ is convergent.

Alternative. Limit comparison test (LCT): Suppose $a_k, b_k > 0$ for all $k \ge 1$, and $\lim\left(\frac{a_k}{b_k}\right) = L \ne 0$. Then the series $\sum_{k=1}^{\infty} a_k$ converges if and only if $\sum_{k=1}^{\infty} b_k$ converges.

(a) $\sum_{k=1}^{\infty} \frac{k^2}{k^3+1}$ diverges because $\lim\left(\frac{k^2}{k^3+1} \div \frac{1}{k}\right) = \lim\left(\frac{k^3}{k^3+1}\right) = 1 \ne 0$ and $\sum_{k=1}^{\infty} \frac{1}{k}$ diverges.

(b) $\sum_{k=1}^{\infty} \frac{k}{k^3+1}$ converges because $\lim\left(\frac{k}{k^3+1} \div \frac{1}{k^2}\right) = \lim\left(\frac{k^3}{k^3+1}\right) = 1 \ne 0$ and $\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges.

Question 5. Comment on the convergence/divergence of the following series.

(a) $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k}}$ (b) $\sum_{k=1}^{\infty} \frac{1}{k!}$ (c) $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt[3]{k}}$ (d) $\sum_{k=1}^{\infty} \frac{(-1)^k k}{k+7}$ (e) $\sum_{k=1}^{\infty} \left(\sqrt{k+1} - \sqrt{k}\right)$

(f) $\sum_{k=3}^{\infty} \frac{\ln k}{\ln(\ln k)}$ (g) $\sum_{k=1}^{\infty} \left(\frac{2}{k} + \frac{3}{k^2}\right)$ (h) $\sum_{k=1}^{\infty} \frac{\sin\left(\frac{k\pi}{2}\right)}{k}$

Solution. Alternating series test (AST): $\sum_{k=1}^{\infty} (-1)^k a_k$ converges if $\left(a_k\right)$ is monotone decreasing and $\lim\left(a_k\right) = 0$.

(a) $\left(a_k\right) = \left(\frac{1}{\sqrt{k}}\right)$ decreases ($k+1 > k > 0 \Rightarrow \sqrt{k+1} > \sqrt{k} \Rightarrow \frac{1}{\sqrt{k+1}} < \frac{1}{\sqrt{k}}$) and $\lim\left(\frac{1}{\sqrt{k}}\right) = 0$.

By AST, $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k}}$ converges. $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k}}$ converges conditionally because $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k}}$ diverges.

(b) $k! \geq \frac{1}{2}k^2 > 0 \Rightarrow 0 < \frac{1}{k!} \leq \frac{2}{k^2}\ \forall k \in \mathbb{N}$ and $\sum_{k=1}^{\infty}\frac{2}{k^2}$ converges, by CT, $\sum_{k=1}^{\infty}\frac{1}{k!}$ converges.

(c) $\left(a_k\right) = \left(\frac{1}{\sqrt[3]{k}}\right)$ decreases and $\lim\left(\frac{1}{\sqrt[3]{k}}\right) = 0$. By AST, $\sum_{k=1}^{\infty}\frac{(-1)^k}{\sqrt[3]{k}}$ converges. Note that $\sum_{k=1}^{\infty}\frac{(-1)^k}{\sqrt[3]{k}}$ converges conditionally because $\sum_{k=1}^{\infty}\frac{1}{\sqrt[3]{k}}$ diverges.

(d) $\lim\left(\frac{(-1)^k k}{k+7}\right) = \lim\left((-1)^k(1-\frac{7}{k+7})\right) \nexists\ \forall k \in \mathbb{N}$ and by DT, $\sum_{k=1}^{\infty}\frac{(-1)^k k}{k+7}$ diverges.

(e) The partial sum $\sum_{k=1}^{n}\left(\sqrt{k+1}-\sqrt{k}\right) = \sqrt{n+1}-1 \to \infty$ as $n \to \infty$ (telescoping). Hence, $\sum_{k=1}^{\infty}\left(\sqrt{k+1}-\sqrt{k}\right)$ diverges. Now $\sum_{k=1}^{\infty}(-1)^k\left(\sqrt{k+1}-\sqrt{k}\right) = \sum_{k=1}^{\infty}\frac{(-1)^k}{\sqrt{k+1}+\sqrt{k}}$. Then $\frac{1}{\sqrt{k+1}+\sqrt{k}} \to 0$ as $k \to \infty$ and $\frac{1}{\sqrt{k+1}+\sqrt{k}}$ decreases (prove it!). So $\sum_{k=1}^{\infty}\frac{(-1)^k}{\sqrt{k+1}+\sqrt{k}}$ converges (but not converges absolutely). Hence, the series $\sum_{k=1}^{\infty}\frac{(-1)^k}{\sqrt{k+1}+\sqrt{k}}$ converges conditionally.

(f) Note that $x > \ln x > 0\ \forall x \in \mathbb{N} \Rightarrow \ln x > \ln(\ln x) > 0\ \forall x \in \mathbb{N}$. We first prove that $\lim\left(\frac{\ln k}{\ln(\ln k)}\right) \neq 0$. Let $\lim\left(\frac{\ln k}{\ln(\ln k)}\right) = 0$. Then $\forall \varepsilon > 0\ \exists N \in \mathbb{N} \ni k > N > 2 \Rightarrow \left|\frac{\ln k}{\ln(\ln k)}\right| < \varepsilon$. Thus, $-\varepsilon < \frac{\ln k}{\ln(\ln k)} < \varepsilon$. Pick $\varepsilon = 1$, then $\ln k < \ln(\ln k)$, a contradiction. Hence, by DT, $\sum_{k=3}^{\infty}\frac{\ln k}{\ln(\ln k)}$ diverges. Alternative: $\lim_{x\to\infty}\frac{\ln x}{\ln(\ln x)} = \lim_{x\to\infty}\frac{\frac{1}{x}}{\frac{1}{x\ln x}} = \lim_{x\to\infty}\ln x = \infty$ (L'Hôpital's rule).

(g) $a_k = \frac{2}{k}+\frac{3}{k^2} > \frac{2}{k}\ \forall k \in \mathbb{N}$. By CT, $\sum_{k=1}^{\infty}\left(\frac{2}{k}+\frac{3}{k^2}\right)$ diverges since $2\sum_{k=1}^{\infty}\frac{1}{k}$ diverges.

Alternative: By LCT, $\sum_{k=1}^{\infty}\left(\frac{2}{k}+\frac{3}{k^2}\right)$ diverges because $\sum_{k=1}^{\infty}\frac{1}{k}$ diverges and $\lim\left(\left(\frac{2}{k}+\frac{3}{k^2}\right) \div \frac{1}{k}\right) = \lim\left(2+\frac{3}{k}\right) = 2 \neq 0$.

Alternative: Since convergent series + divergent series = divergent series, we have divergent series $= \sum_{k=1}^{\infty}\left(\frac{2}{k}+\frac{3}{k^2}\right) = \sum_{k=1}^{\infty}\frac{2}{k}+\sum_{k=1}^{\infty}\frac{3}{k^2}$ = divergent series + convergent series.

(h) $\sum_{k=1}^{\infty}\frac{\sin\left(\frac{k\pi}{2}\right)}{k} = \sum_{r=1}^{\infty}\frac{\sin\left(\frac{(2r-1)\pi}{2}\right)}{2r-1} = \sum_{r=1}^{\infty}\frac{(-1)^{r+1}}{2r-1} = -\sum_{r=1}^{\infty}(-1)^r a_r$, with $a_r = \frac{1}{2r-1}$. Then we see that $\left(a_k\right)$ is monotonic decreasing and $\lim\left(a_k\right) = 0$. Hence, the series converges by AST.

Question 6. Prove Theorem 3.5 of Chapter 3.

Let $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ be convergent series with $\sum_{k=1}^{\infty} a_k = \alpha$ and $\sum_{k=1}^{\infty} b_k = \beta$. Then

(a) $\sum_{k=1}^{\infty} \left(ca_k\right) = c\alpha$ and (b) $\sum_{k=1}^{\infty} \left(a_k + b_k\right) = \alpha + \beta$, where $c, \alpha, \beta \in \mathbb{R}$ are constants.

Proof. (a) Let $s_n = \sum_{k=1}^{n} a_k$. Given $\sum_{k=1}^{\infty} a_k = \alpha$, meaning that $\lim\left(s_n\right) = \alpha$. Applying the Algebraic Limit Theorem for sequences (Theorem 6.1, Chapter 2) yields

$$\lim\left(cs_n\right) = \sum_{k=1}^{\infty} ca_k = c\alpha, \text{ as desired.}$$

(b) Let $s_n = \sum_{k=1}^{n} a_k$ and $t_n = \sum_{k=1}^{n} b_k$. Given that $\left(s_n\right)$, $\left(t_n\right)$ converge to α and β, respectively. By the Theorem 6.1 of Chapter 2, $\left(s_n + t_n\right)$ converge to $\alpha + \beta$. Hence,

$$\lim\left(s_n + t_n\right) = \sum_{k=1}^{\infty} \left(a_k + b_k\right) = \alpha + \beta.$$

Question 7. Prove the statement $\left|s_n - s_m\right| \le \left|a_{m+1}\right| \ \forall n > m$ in Theorem 4.9 of Chapter 3. Given that $\left(a_k\right)$ is decreasing and $\lim\left(a_k\right) = 0 \ \forall k \in \mathbb{N}$.

Proof.
$$\begin{aligned}
\left|s_n - s_m\right| &= \left|(-1)^{m+1} a_{m+1} + (-1)^{m+2} a_{m+2} + (-1)^{m+3} a_{m+3} + \cdots + (-1)^n a_n\right| \\
&= \left|(-1)^{m+1}\right| \left|a_{m+1} - a_{m+2} + a_{m+3} - \cdots + (-1)^{n-m-1} a_n\right| \\
&= \left|a_{m+1} - a_{m+2} + a_{m+3} - \cdots + (-1)^{n-m-1} a_n\right|
\end{aligned}$$

Since $\left(a_k\right)$ is decreasing and $\lim\left(a_k\right) = 0$, it follows that $a_k \ge 0 \ \forall k \in \mathbb{N}$ (Q15(a)).

If $(-1)^{n-m-1} a_n = -a_n$, then by the fact that $\left(a_n\right)$ is monotone decreasing,

$$\begin{aligned}
\left|s_n - s_m\right| &= \left|a_{m+1} - a_{m+2} + a_{m+3} - a_{m+4} + a_{m+5} \cdots - a_{n-2} + a_{n-1} - a_n\right|. \\
&\le \left|a_{m+1} - a_{m+2} + a_{m+2} - a_{m+4} + a_{m+4} \cdots - a_{n-2} + a_{n-2} - a_n\right| \\
&= \left|a_{m+1} - a_n\right| \\
&< \left|a_{m+1}\right|
\end{aligned}$$

If $(-1)^{n-m-1} a_n = a_n$, then by the fact that $\left(a_n\right)$ is monotone decreasing,

$$\begin{aligned}
\left|s_n - s_m\right| &= \left|a_{m+1} - a_{m+2} + a_{m+3} - \cdots - a_{n-1} + a_n\right| \\
&\le \left|a_{m+1} - a_{m+2} + a_{m+2} - \cdots - a_{n-1} + a_{n-1}\right| \\
&= \left|a_{m+1}\right|
\end{aligned}$$

Question 8. Prove that $\sum_{k=1}^{\infty} a_k$ diverges implies that $\sum_{k=1}^{\infty} ca_k$ diverges, $c \in \mathbb{R} \setminus \{0\}$.

Proof. Let $c \in \mathbb{R} \setminus \{0\}$. By Cauchy Criterion for Series, $\sum_{k=1}^{\infty} ca_k$ converges

$\Leftrightarrow \forall \tilde{\varepsilon} = |c|\varepsilon > 0 \ \exists N \in \mathbb{N}$ such that $n > m > N \Rightarrow \left|\sum_{k=m+1}^{n} ca_k\right| < \tilde{\varepsilon} = |c|\varepsilon \ \forall n, m \in \mathbb{N}$

$\Leftrightarrow \forall \varepsilon > 0 \ \exists N \in \mathbb{N}$ such that $n > m > N \Rightarrow \left|\sum_{k=m+1}^{n} a_k\right| < \varepsilon \ \forall n, m \in \mathbb{N} \Leftrightarrow \sum_{k=1}^{\infty} a_k$ converges.

Remark: The result is true in both directions.

Question 9. The n-th root test (NRT): Let $\sum_{k=1}^{\infty} a_k$, $a_k \geq 0 \ \forall k \in \mathbb{N}$ and $L = \lim\left(\sqrt[k]{a_k}\right)$.

(a) Let $L < 1$ and $\varepsilon = \frac{1-L}{2} > 0$. Prove that $\exists N_1 \in \mathbb{N}$ such that

$$(a_k)^{\frac{1}{k}} < L + \varepsilon \quad \forall k > N_1.$$

(b) Using part (a), prove that $\sum_{k=1}^{\infty} a_k$ converges if $L < 1$.

(c) Let $L > 1$ and $\varepsilon = \frac{L-1}{2} > 0$. Prove that $\exists N_2 \in \mathbb{N}$ such that

$$(a_k)^{\frac{1}{k}} > L - \varepsilon \quad \forall k > N_2.$$

(d) Using part (c), prove that $\sum_{k=1}^{\infty} a_k$ diverges if $L > 1$.

Proof. $\forall \varepsilon > 0, \exists N \in \mathbb{N} \ni \left|\sqrt[k]{a_k} - L\right| < \varepsilon \ \forall k > N$. Note that $L \geq 0$.

(a) Suppose $L < 1$ and $\varepsilon = \frac{1-L}{2} > 0$. Then $\exists N_1 \in \mathbb{N}$ such that

$$\left|a_k^{\frac{1}{k}} - L\right| < \varepsilon \Leftrightarrow L - \varepsilon < a_k^{\frac{1}{k}} < L + \varepsilon \ \forall k > N_1.$$

(b) $0 \leq (a_k)^{\frac{1}{k}} < L + \varepsilon = L + \frac{1-L}{2} = \frac{L+1}{2} < 1 \Rightarrow 0 \leq a_k < \left(\frac{L+1}{2}\right)^k < 1 \ \forall k > N_1$ (since $0 < x < 1$ $\Rightarrow 0 < x^n < 1 \ \forall x \in \mathbb{R}$). Now $\sum_{k=1}^{\infty} \left(\frac{L+1}{2}\right)^k$ is a convergent geometry series, it follows by CT that $\sum_{k=N_1+1}^{\infty} a_k$ converges and hence $\sum_{k=1}^{\infty} a_k$ converges.

(c) Suppose $L > 1$. Define $\varepsilon = \frac{L-1}{2} > 0$. Then $\exists N_2 \in \mathbb{N}$ such that

$$\left|a_k^{\frac{1}{k}} - L\right| < \varepsilon \Leftrightarrow L - \varepsilon < a_k^{\frac{1}{k}} < L + \varepsilon \ \forall k > N_2.$$

(d) Note that $1 < \frac{L+1}{2} = L - \frac{L-1}{2} = L - \varepsilon < (a_k)^{\frac{1}{k}}$. Thus, $a_k > \left(\frac{L+1}{2}\right)^k > 1 \ \forall k > N_2$.

By the Theorem 6.3 (Chap 2) and the DT, $\lim\left(a_k\right) \geq 1 \neq 0$ and hence the series $\sum_{k=1}^{\infty} a_k$ diverges if $L > 1$.

Question 10.

(a) Prove that if $\left(ka_k\right)$ converges to a nonzero real number, then $\sum_{k=1}^{\infty} a_k$ diverges. Give a counter example to prove the converse is false.

(b) Prove that if $\left(k^2 a_k\right)$ converges, then $\sum_{k=1}^{\infty} a_k$ converges. Give a counter example to prove the converse is false.

Proof. (a) Let $\lim\left(ka_k\right) = a \neq 0$. So $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $k > N \Rightarrow \left|ka_k - a\right| < \varepsilon$. Rewrite the inequality we have $a - \varepsilon < ka_k < a + \varepsilon$.

If $a > 0$. Then set $\varepsilon = \frac{a}{2} > 0$ and we have $\frac{a}{2} < ka_k \Rightarrow a_k > \frac{a}{2k} > 0$ for all $k > N$. Since $\sum_{k=1}^{\infty} \frac{a}{2k} = \frac{a}{2}\sum_{k=1}^{\infty} \frac{1}{k}$ is divergent, it follows by CT that $\sum_{k=1}^{\infty} a_k$ is divergent.

If $a < 0$. Then set $\varepsilon = -\frac{a}{2} > 0$ and we have $ka_k < \frac{a}{2} < 0 \Rightarrow -a_k > -\frac{a}{2k} > 0 \quad \forall k > N$. Since $\sum_{k=1}^{\infty} -\frac{a}{2k} = -\frac{a}{2}\sum_{k=1}^{\infty} \frac{1}{k}$ is divergent, it follows by CT that $\sum_{k=1}^{\infty}\left(-a_k\right)$ is divergent. By Theorem 3.5, $\sum_{k=1}^{\infty} a_k$ is divergent.

The converse is false, for example $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k}}$ diverges, but $\left(k\left(\frac{1}{\sqrt{k}}\right)\right) = \left(\sqrt{k}\right)$ diverges.

(b) Let $\lim\left(k^2 a_k\right) = L$. Then $\forall \varepsilon > 0, \exists N \in \mathbb{N} \ni \left|k^2 a_k - L\right| < \varepsilon \ \forall k > N$.

If $L > 0$. Set $\varepsilon = L$, then $0 < k^2 a_k < 2L \Rightarrow 0 < a_k < \frac{2L}{k^2}$. So, $\sum_{k=1}^{\infty} a_k$ converges by CT since $2L\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges.

If $L < 0$. Set $\varepsilon = -L$, then $2L < k^2 a_k < 0 \Rightarrow 0 < -a_k < -\frac{2L}{k^2}$. So, $\sum_{k=1}^{\infty} a_k$ converges by CT (since $-2L\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges) and Theorem 3.5.

If $L = 0$. Then $0 < \left|a_k\right| < \frac{\varepsilon}{k^2}$. So, $\sum_{k=1}^{\infty} \left|a_k\right|$ converges by CT (since $\varepsilon\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges) and Theorem 3.5. Then by ACT, $\sum_{k=1}^{\infty} a_k$ converges.

The converse is false, for example, $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k^3}}$ converges but $\left(k^2 \cdot \frac{1}{\sqrt{k^3}}\right) = \left(\sqrt{k}\right)$ diverges.

Question 11. Give an example of a divergent series $\sum_{k=1}^{\infty} a_k$ and a convergent series $\sum_{k=1}^{\infty} b_k$ such that $a_k \le b_k \ \forall k \in \mathbb{N}$.

Answer. $\sum_{k=1}^{\infty} \frac{1}{k^2}$ converges and $-\sum_{k=1}^{\infty} \frac{1}{k}$ diverges, and $\frac{1}{k^2} \ge -\frac{1}{k} \ \forall k \in \mathbb{N}$.

Alternative: Choose $a_k = 0$ and $b_k = -1$.

Question 12. Discuss the convergence/divergence/absolute/conditional convergence of the series: (a) $\sum_{k=1}^{\infty} (-1)^k \frac{\sqrt{k+1}-\sqrt{k}}{k^p}, p > 0$ (b) $\sum_{k=1}^{\infty} \frac{1}{p^k - q^k}, p > q > 0$ (c) $\sum_{k=1}^{\infty} \frac{1}{p^k}, p \in \mathbb{R} \setminus \{0\}$

Solution. (a) Note that $\frac{\sqrt{k+1}-\sqrt{k}}{k^p} = \frac{1}{k^p\left(\sqrt{k+1}+\sqrt{k}\right)} \to 0$ and decreasing. Hence $\sum_{k=1}^{\infty} (-1)^k \frac{\sqrt{k+1}-\sqrt{k}}{k^p}$ converges. Next, note that $0 < \frac{1}{2k^{p+\frac{1}{2}}} = \frac{1}{2k^p\sqrt{2k}} < \frac{1}{k^p\left(\sqrt{k+1}+\sqrt{k}\right)} < \frac{1}{k^p\sqrt{k}} = \frac{1}{k^{p+\frac{1}{2}}}$. Hence, by CT, $\sum_{k=1}^{\infty} (-1)^k \frac{\sqrt{k+1}-\sqrt{k}}{k^p}$ converges absolutely if $p > \frac{1}{2}$ and converges conditionally if $p \le \frac{1}{2}$.

(b) Given $p > q > 0$. Thus, $0 < p^k - q^k < p^k \Rightarrow \frac{1}{p^k - q^k} > \frac{1}{p^k}$.

By GS and CT, $\sum_{k=1}^{\infty} \frac{1}{p^k - q^k}$ diverges if $p \le 1$ (since $\sum_{k=1}^{\infty} \frac{1}{p^k}$ diverges when $\frac{1}{p} \ge 1$).

Next, note that $L = \lim\left(\frac{p^k - q^k}{p^{k+1} - q^{k+1}}\right) = \frac{1}{p}\lim\left(\frac{1-\left(\frac{q}{p}\right)^k}{1-\left(\frac{q}{p}\right)^{k+1}}\right) = \frac{1}{p}$. By ratio test, the series converges if $p > 1$ and diverges if $p < 1$.

(c) $\sum_{k=1}^{\infty} \frac{1}{p^k}$ converges if $\left|\frac{1}{p}\right| < 1 \Leftrightarrow |p| > 1$ and diverges if $\left|\frac{1}{p}\right| \ge 1 \Leftrightarrow |p| \le 1, p \ne 0$.

In particular, if $p > 1$, then $\sum_{k=1}^{\infty} \left|\frac{1}{p^k}\right| = \sum_{k=1}^{\infty} \frac{1}{p^k}$ converges absolutely.

If $p < -1$, then write $\sum_{k=1}^{\infty} \frac{1}{p^k} = \sum_{k=1}^{\infty} \frac{(-1)^k}{q^k}$, where $q = -p > 1$.

Thus, $\sum_{k=1}^{\infty} \left|\frac{1}{p^k}\right| = \sum_{k=1}^{\infty} \left|\frac{(-1)^k}{q^k}\right| = \sum_{k=1}^{\infty} \frac{1}{q^k}$ is converges because $\frac{1}{q} < 1$ (since $-p = q > 1$),

which means $\sum_{k=1}^{\infty} \frac{1}{p^k}$ converges absolutely for $|p| > 1$ and diverges when $|p| \le 1, p \ne 0$.

The series does not converge conditionally for any range of p.

Question 13. Let $\left(a_k\right)$ and $\left(b_k\right)$ be two sequences such that $a_k = b_{k+1} - b_k \ \forall k \in \mathbb{N}$. Prove that $\sum_{k=1}^{\infty} a_k$ converges if and only if $\lim\left(b_k\right)$ exists, and additionally show that

$$\sum_{k=1}^{\infty} a_k = \lim\left(b_k\right) - b_1 .$$

Proof. Let $s_n = \sum_{k=1}^{n} a_k$. Then $s_n = \sum_{k=1}^{n} (b_{k+1} - b_k) = b_{n+1} - b_1$ (telescoping series). Hence,

$$\sum_{k=1}^{\infty} a_k \text{ converges} \Leftrightarrow \left(s_n\right) \text{ converges} \Leftrightarrow \lim\left(b_k\right) \text{ exists.}$$

Suppose $\lim\left(b_k\right)$ exists, then $\sum_{k=1}^{\infty} a_k = \lim\left(s_n\right) = \lim\left(b_k\right) - b_1$.

Question 14. Prove that if $a_k > 0 \ \forall k \in \mathbb{N}$ and $\sum_{k=1}^{\infty} a_k^2$ converges, then $\sum_{k=1}^{\infty} \frac{a_k}{k}$ converges.

Proof. Applying the Cauchy-Schwarz inequality: $\left(\sum_{k=1}^{n} x_k y_k\right)^2 \le \left(\sum_{k=1}^{n} x_k^2\right)\left(\sum_{k=1}^{n} y_k^2\right)$. Thus,

$$\left(\sum_{k=1}^{n} \frac{a_k}{k}\right)^2 \le \left(\sum_{k=1}^{n} a_k^2\right)\left(\sum_{k=1}^{n} \frac{1}{k^2}\right) \Rightarrow 0 < \sum_{k=1}^{n} \frac{a_k}{k} \le \sqrt{\left(\sum_{k=1}^{n} a_k^2\right)\left(\sum_{k=1}^{n} \frac{1}{k^2}\right)} \quad \because a_k > 0 \ \forall k \in \mathbb{N} .$$

Let $s_n = \sum_{k=1}^{n} \frac{a_k}{k}$, $t_n = \sum_{k=1}^{n} a_k^2$, $u_n = \sum_{k=1}^{n} \frac{1}{k^2}$. So, $0 < s_n \le \sqrt{t_n u_n}$. Note that $\left(s_n\right)$ increases; and bounded above if $\left(\sqrt{t_n u_n}\right)$ converges. Now since both $\left(t_n\right)$,$\left(u_n\right)$ converges, it follows that $\left(\sqrt{t_n u_n}\right)$ is convergent (see below*). Hence, by MCT, $\left(s_n\right)$ is convergent $\sum_{k=1}^{\infty} \frac{a_k}{k}$ converges.

Alternative: First, we prove that if $\sum_{r=1}^{\infty} a_r^2$ and $\sum_{r=1}^{\infty} b_r^2$ converge, a_r,$b_r \ge 0$ for all $r \ge 1$. Then $\sum_{r=1}^{\infty} a_r b_r$ is convergent. From the fact $(a_r - b_r)^2 \ge 0 \Rightarrow \frac{1}{2}\left(a_r^2 + b_r^2\right) \ge a_r b_r$, and since $\frac{1}{2}\sum_{r=1}^{\infty}\left(a_r^2 + b_r^2\right)$ converges, it follows by the comparison test that $\sum_{r=1}^{\infty} a_r b_r$ converges. Now let $b_r = \frac{1}{r}$. Then $\sum_{r=1}^{\infty} b_r^2$ converges. Given that $\sum_{r=1}^{\infty} a_r^2$ converges, so is $\sum_{r=1}^{\infty} a_r b_r = \sum_{r=1}^{\infty} \frac{a_r}{r}$.

*Let $\left(a_n\right)$ be a converges sequence with $a_n \ge 0 \ \ \forall n \in \mathbb{N}$. Then $\lim\left(a_n\right) = a$ implies that $\lim\left(\sqrt{a_n}\right) = \sqrt{a}$.

Proof. By Theorem 6.3, $a_n \ge 0 \ \forall n \in \mathbb{N} \Rightarrow \lim\left(a_n\right) = a \ge 0$. So, $\sqrt{a}$ is defined.

Case 1. Suppose $\exists N \in \mathbb{N} \ni a_n > 0$ for $n > N$ (finitely many zero elements in the sequence). Then from the definition $\lim(a_n) = a$ $\forall \varepsilon\sqrt{a} > 0$, $\exists N \in \mathbb{N} \ni |a_n - a| < \varepsilon\sqrt{a}$ for $n > N$. Note that $\sqrt{a_n} + \sqrt{a} > \sqrt{a}$ for $n > N$. Thus, $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that

$$\left|\sqrt{a_n} - \sqrt{a}\right| = \left|\frac{a_n - a}{\sqrt{a_n} + \sqrt{a}}\right| < \frac{1}{\sqrt{a}}|a_n - a| < \varepsilon \text{ whenever } n > N.$$

Case 2. Suppose $\exists N \in \mathbb{N} \ni a_n = 0$ for $n > N$ (infinitely many zero elements in the sequence). Then $\sqrt{a_n} = 0$ for $n > N$. We claim that $\lim(a_n) = \lim\left(\sqrt{a_n}\right) = 0$. For, $|a_n - 0| = \left|\sqrt{a_n} - 0\right| = 0 < \varepsilon$ whenever $n > N$.

Combining Case 1 and Case 2, we have $\lim\left(\sqrt{a_n}\right) = \sqrt{a} \geq 0$.

Question 15.

(a) Let (a_n) be a decreasing sequence with $\lim(a_n) = 0$. Prove that $a_n \geq 0$ $\forall n \in \mathbb{N}$.

(b) Hence, prove that if (a_n) decreases and $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim(na_n) = 0$.

Proof. (a) Suppose $\exists m \in \mathbb{N} \ni a_m < 0$. Given that $\lim(a_n) = 0$. Thus,

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } |a_n| < \varepsilon \ \forall n > N.$$

Let $\varepsilon = -\frac{a_m}{2} > 0$. Then $\frac{a_m}{2} < a_k < -\frac{a_m}{2} \Rightarrow m \leq N$. (If $m > N$, then we can take $n = m$ which implies that $a_m > \frac{a_m}{2} \Rightarrow a_m > 0$, violating $a_m < 0$). In other words, $a_m < 0$ exist happen for finitely many times and $m \leq N$. Given that the sequence is decreasing, and since $a_m < 0$, it follows that $0 > a_m \geq a_{m+1} \geq \cdots \geq a_N \geq a_{N+1} \geq \cdots$, which means $a_{N+1} < 0$, a contradiction.

(b) Given that $\sum_{k=1}^{\infty} a_k$ is convergent, which implies that $\lim(a_n) = 0$. So, from (a), $a_n \geq 0$ $\forall n \in \mathbb{N}$ since (a_n) is decreasing. Let $\varepsilon > 0$. Then from the Cauchy Criterion, there exists an $M \in \mathbb{N}$ such that for all $n > M$,

$$|a_{M+1} + a_{M+2} + \cdots + a_n| = a_{M+1} + a_{M+2} + \cdots + a_n < \tfrac{\varepsilon}{2} \text{ (since } a_n \geq 0 \ \forall n \in \mathbb{N}).$$

Since $\lim(a_n) = 0$, it follows that $\forall \frac{\varepsilon}{2M} > 0, \exists P \in \mathbb{N}$ such that $n > P \Rightarrow |a_n| = a_n < \frac{\varepsilon}{2M}$. Now let $N = \max\{M, P\}$. Then $n > N$ implies that

$$\begin{aligned} na_n &= a_n + a_n + \cdots + a_n + a_n + a_n + \cdots + a_n \\ &\leq \underbrace{a_P + a_P + \cdots + a_P}_{M \text{ terms}} + \underbrace{a_{M+1} + a_{M+2} + \cdots + a_n}_{n-M \text{ terms}} \text{ (decreasing sequence)} \\ &< Ma_P + \tfrac{\varepsilon}{2} < M\left(\tfrac{\varepsilon}{2M}\right) + \tfrac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Question 16. Let $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ be two series. Then $\sum_{k=1}^{\infty} b_k$ is called a rearrangement of $\sum_{k=1}^{\infty} a_k$ if there exists a bijective function $\sigma : \mathbb{N} \to \mathbb{N}$ such that $b_{\sigma(k)} = a_k$.

(a) Prove that if a series converges absolutely, then any rearrangement of this series converges to the same limit.

(b) Prove that if $\sum_{k=1}^{\infty} a_k$ diverges to positive infinity with $a_k \geq 0 \ \forall k \in \mathbb{N}$, then any rearrangement of this series diverges to positive infinity.

Proof. Let $s_n = \sum_{k=1}^{n} a_k$ and $t_m = \sum_{k=1}^{m} b_k$ be partial sums of $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$. Suppose $\sum_{k=1}^{\infty} b_k$ is a rearrangement of $\sum_{k=1}^{\infty} a_k$.

(a) Given $\sum_{k=1}^{\infty} a_k$ converges absolutely to a. We prove that $\lim\left(t_m\right) = a$. Since $\sum_{k=1}^{\infty} \left|a_k\right|$ converges to a, it follows that $\sum_{k=1}^{\infty} a_k$ converges to a and so $\lim\left(s_n\right) = a$. By the definition of $\lim\left(s_n\right) = a$, take $\frac{\varepsilon}{2} > 0$ and choose $N_1 \in \mathbb{N}$ such that

$$\left|s_n - a\right| < \tfrac{\varepsilon}{2} \forall n > N_1 .$$

By the Cauchy Criterion for series, take $\frac{\varepsilon}{2} > 0$ and choose $N_2 \in \mathbb{N}$ such that

$$\sum_{k=m+1}^{n} \left|a_k\right| < \tfrac{\varepsilon}{2} \forall n > m > N_2 .$$

Now, take $N = \max\left\{N_1, N_2\right\}$. We know that all the finite elements $\left\{a_i : i = 1,...,N\right\}$ must appear in the rearranged series, and we want to move far enough out in $\sum_{k=1}^{\infty} b_k$ so that we have included all of these terms. So, choose $M = \max\left\{\sigma(1), \sigma(2),..., \sigma(N)\right\}$. It is clear now that if $m \geq M$, then sequence $\left(t_m - s_N\right)$ consists of a finite set of terms, the absolute values of which appear in the tail $\sum_{k=N+1}^{\infty} \left|a_k\right|$. The choice of N_2 earlier then guarantees $\left|t_m - s_N\right| < \frac{\varepsilon}{2}$, and so

$$\left|t_m - a\right| = \left|t_m - s_N + s_N - a\right| \leq \left|t_m - s_N\right| + \left|s_N - a\right| < \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon \text{ whenever } m \geq M .$$

(b) Given that $\lim\left(s_n\right) = \infty$. Then $\forall L > 0, \exists N \in \mathbb{N} \ni s_n > L$ whenever $n > N$. We shall prove that $\lim\left(t_m\right) = \infty$. Choose

$$M = \max\left\{\sigma(1), \sigma(2),..., \sigma(N)\right\} \geq N .$$

Then $t_m > L$ whenever $m > M$.

Question 17. Let $\rho : \mathbb{N} \to \mathbb{N}$ be a bijective function such that $\rho(k) < \rho(l)$ whenever $k < l$. Let $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ be two series. Then $\sum_{k=1}^{\infty} b_k$ is said to be obtained from $\sum_{k=1}^{\infty} a_k$ by **inserting parentheses** if $b_1 = \sum_{k=1}^{\rho(1)} a_k$ and $b_{k+1} = \sum_{k=\rho(k)+1}^{\rho(k+1)} a_k \forall k \in \mathbb{N}$. Equivalently, it could also be said that $\sum_{k=1}^{\infty} a_k$ is obtained from $\sum_{k=1}^{\infty} b_k$ by **removing parentheses**.

(a) Prove that if $\sum_{k=1}^{\infty} a_k$ is convergent series that converges to a and $\sum_{k=1}^{\infty} b_k$ is obtained from $\sum_{k=1}^{\infty} a_k$ by inserting parentheses, then the series $\sum_{k=1}^{\infty} b_k$ converges to a.

(b) Prove or disprove the converse of part (a).

Proof. (a) Let $\sum_{k=1}^{\infty} a_k$ be convergent series that converges to a. Then by the Cauchy Criterion, for all $\varepsilon > 0$ there exists an $N_1 \in \mathbb{N}$ such that

$$\left|a_{m+1} + a_{m+2} + \cdots + a_n\right| < \varepsilon \text{ whenever } n > m > N_1.$$

Since $\rho(k)$ is a monotone increasing function, it follows that $a_{\rho(k)} \geq a_k$ for all $k \in \mathbb{N}$.

So, $\left|b_{m+1} + b_{m+2} + \cdots + b_n\right|$

$$= \left|\left(a_{\rho(m)+1} + \cdots + a_{\rho(m+1)}\right) + \left(a_{\rho(m+1)+1} + \cdots + a_{\rho(m+2)}\right) + \cdots + \left(a_{\rho(n-1)+1} + \cdots + a_{\rho(n)}\right)\right| < \varepsilon.$$

Hence, Cauchy's condition is satisfied for the series $\sum_{k=1}^{\infty} b_k$, so $\sum_{k=1}^{\infty} b_k$ converges.

All that remains to show is that $\sum_{k=1}^{\infty} b_k$ converges to a.

Let $s_n = \sum_{k=1}^{n} a_k$ and $t_m = \sum_{k=1}^{m} b_k$ be partial sums of $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$. Then $\left(s_n\right)$ converges to a, and since $\left(t_m\right)$ is a subsequence of $\left(s_n\right)$ we must have that $\left(t_m\right)$ also converges to a. Hence, $\sum_{k=1}^{\infty} b_k$ converges to a.

(b) The converse of (a) is not true in general, i.e., if $\sum_{k=1}^{\infty} b_k$ is obtained by inserting parentheses into $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ converges, then $\sum_{k=1}^{\infty} a_k$ may converge or diverge.

A counter example: Let $\sum_{k=1}^{\infty} a_k = \sum_{k=1}^{\infty} (-1)^k$. This series clearly diverges. If we insert parentheses by letting $b_k = a_{2k-1} + a_{2k}\ \forall k \in \mathbb{N}$. Then we have that

$$b_k = (-1)^{2k-1} + (-1)^{2k} = 0\ \forall k \in \mathbb{N}\,.$$

So $\sum_{k=1}^{\infty} b_k$, which is obtained by inserting parentheses is convergent.